ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

Automated Phissing Email Detection and Reporting System

SUT PROJECTS

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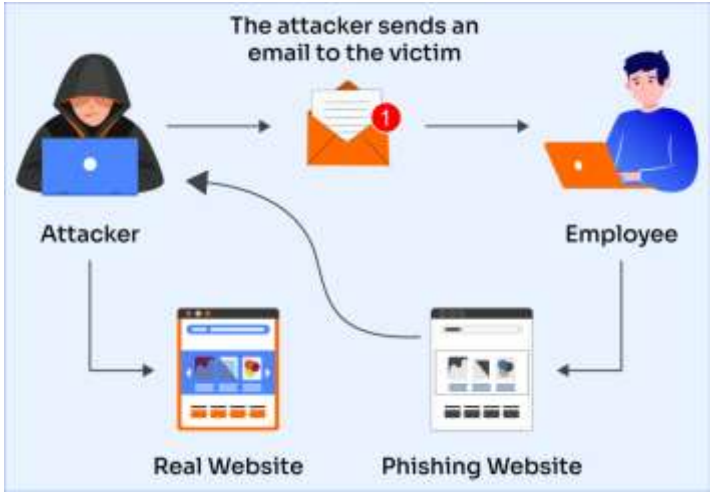
Introduction

Problem Statement

Phishing Attack:

- Phishing is a **cyber attack** where attackers **trick users** into giving their sensitive information.
- Attackers send **fake emails, messages, or links** that look real
- They pretend to be **banks, companies, or trusted organizations**
- Victims are asked to **click a link or enter login details**



Problem Statement (cont.)

Phishing attacks target **anyone who uses email or online services**, including:

- **Individuals** (regular users)
- **Employees** in companies
- **Businesses and organizations**
- **Banks and financial institutions**
- **Students and universities**

This can cause **financial loss**, **steal passwords and personal data**, or **disrupting organizations systems**.

Objective: Implementing a phishing email detection system

A phishing email detection system helps **identify and block phishing emails** before they reach users.

- Uses **machine learning and security rules** to analyze emails
- Detects suspicious links, fake senders, and unusual content
- Classifies emails as **Phishing** or **Safe**
- Warns users or automatically blocks malicious emails

Project Idea

- Classify emails as **Safe or Phishing** using content and header of the mail.
- Use a **hybrid approach** combining:
  - Machine Learning
  - NLP techniques
  - Rule-based URL and header checks
- Apply **TF-IDF** for feature extraction from email text.

Project Idea (cont.)

- Use **classical ML algorithms** (Logistic Regression, Naive Bayes, SVM, Random Forest).
- Implement a **multi-stage detection engine** to improve accuracy and robustness.
- Provide **real-time results** with confidence scores and explanations.
- Offer a **web-based interface** with persistent scan logging.
- Deliver a **practical and efficient** phishing detection solution.

Project Motivation

Scientific Perspective

Research Gaps Addressed

- Limited use of **hybrid detection approaches** combining machine learning with rule-based analysis.
- Lack of **explain ability** in many phishing detection systems, reducing trust and usability.
- Poor **generalization to unseen, real-world emails** due to overfitting on benchmark datasets.

Scientific Perspective

Scientific Support & Relevant Studies

- Research shows that **TF-IDF combined with classical ML algorithms** (SVM, Random Forest, Logistic Regression) performs effectively in phishing detection.
- Studies confirm that **URL-based lexical features** significantly improve phishing detection accuracy.

Scientific Perspective

- Literature reviews highlight that **hybrid systems** outperform single-technique models in handling evolving phishing strategies.
- Existing research emphasizes the need for **lightweight, interpretable, and deployable** phishing detection systems for practical use.

Market Perspective

Market Demand & Trends

- Rapid growth in phishing and social engineering attacks.
- Increasing use of AI-generated phishing emails.
- High demand for affordable and easy-to-deploy email security solutions.
- Growing interest in explainable cybersecurity systems.

Impact & Benefits

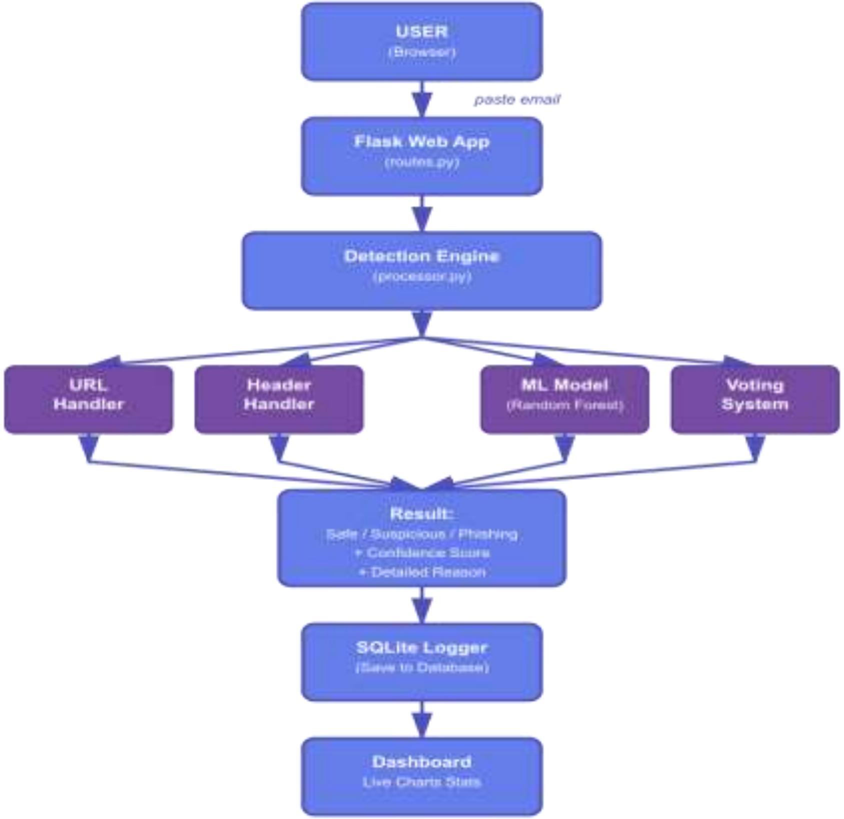
- Reduces phishing-related risks and losses.
- Improves user awareness and trust.
- Provides real-time, transparent email protection.
- Suitable for organizations, universities, and individuals.

Market Perspective

FBI IC3 Reports & APWG Phishing Trends (2020–2024)

System Overview Diagram

System Overview



Similar Systems

Similar systems “comparative studies”

System	Release / Major Update	Detection Accuracy*	Key Approach	Limitations
Barracuda Email Protection	Active AI upgrades (2023–2025)	~99.2%	ML + behavior analysis	Limited explainability, closed system
Microsoft Defender for Office 365	Rebranded 2020, continuous updates	~89–98%	ML + reputation + policies	Weaker against advanced impersonation
Mimecast Email Security	Cloud Integrated expansion (2023)	~99% (overall email security)	Multi-layer AI filtering	Expensive, complex tuning

Barracuda Sentinel / Barracuda Email Protection(2023)

- **Type:** AI-powered email security platform used widely by enterprises(99.2%).
- **Real-World Use:** Trusted by many businesses for proactive protection and automated quarantining.

- **Strengths:**
    - AI pattern learning for internal and external threats
    - Real-time automated remediation
    - Protects against impersonation and account takeover

- **Limitations:**
    - Largely tailored for enterprise environments
    - May require subscription and configuration
    - Might not deeply explain why an email is flagged (visibility is mainly internal)

Microsoft Defender for Office 365 (2021)

- **Type:** Built-in malware and phishing protection for Microsoft 365 customers(89 – 98 %).
- **Real-World Use:** Extensively used by organizations that rely on Microsoft 365, from SMBs to enterprises.

- **Strengths:**
  - Seamless with Microsoft products
  - Automated link/attachment scanning

- **Limitations:**
  - Real users report gaps in advanced spoofing and display-name attacks.
  - Often needs additional layers to fully protect against modern phishing.

Mimecast Email Security(2023)

- **Type:** Cloud-based enterprise email security solution(~98%).
- **Real-World Use:** Commonly deployed in midsize to large businesses.

- **Strengths:**
  - Multi-layered detection (URLs, attachments, content)
  - Integrated awareness training
  - Strong dashboard and reporting

- **Limitations:**
  - Typically more expensive than basic email protection
  - Requires some expertise to tune correctly

What Makes Our System Unique?

- Explainable: Provides clear reasons why an email is classified as phishing
- Detects suspicious URLs and phishing links within emails
- Trained and tested on public benchmark datasets
- Can be adapted to different organizations

Key Takeaway:-

- Our system prioritizes transparency, URL-based threat detection, and adaptability — addressing major limitations of current commercial phishing detection solutions.

Technical Approach

Rule-Based Detection Layer (Chain of Responsibility)

HeaderHandler

- Parses email headers
- Validates sender format
- Detects:
  - Missing or malformed sender
  - Suspicious domains & TLDs
  - Brand impersonation (PayPal, Amazon, Google, etc.)
  - Obfuscated or nested domains

Rule-Based Detection Layer (Chain of Responsibility)

URLHandler:

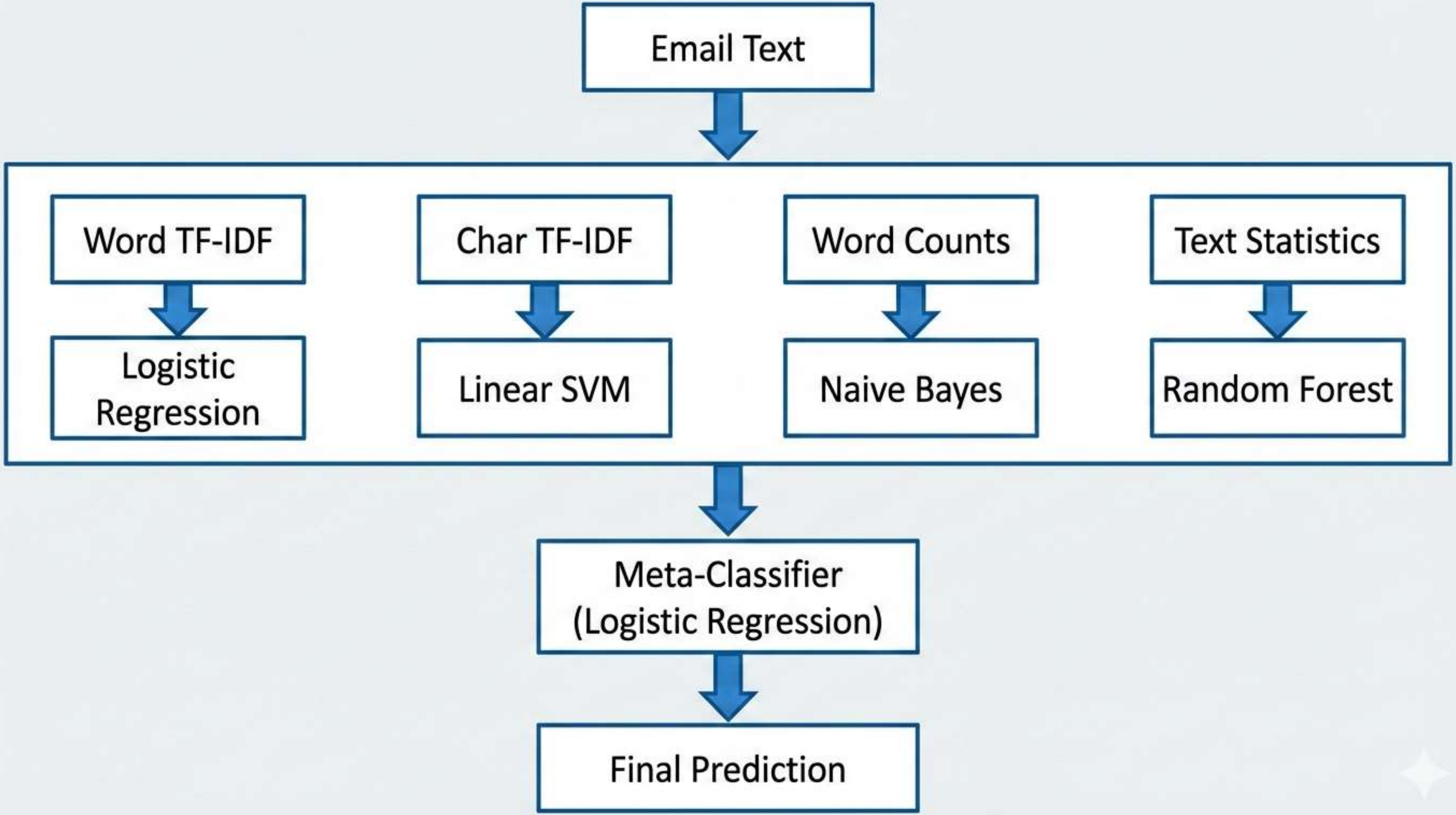
- Extracts URL-based features from email body
- Detects:
  - Long URLs
  - IP-based links
  - URL shorteners
  - Suspicious TLDs
  - Excessive symbols & uppercase
- Computes a **risk score**
- Immediately flags high-risk phishing emails

These layer quickly blocks obvious phishing attempts without using ML, improving speed and reliability.

Ensemble Learning Model Architecture

Main idea:

- Multiple specialized models analyze the same email
- Each model focuses on different phishing signals
- A meta-classifier combines their predictions
- Results in higher accuracy and robustness


Email Text
Word TF-IDF
Logistic Regression
Char TF-IDF
Linear SVM
Word Counts
Naive Bayes
Text Statistics
Random Forest
Meta-Classifier (Logistic Regression)
Final Prediction

Feature Views

- Each view captures **a different phishing behavior**.

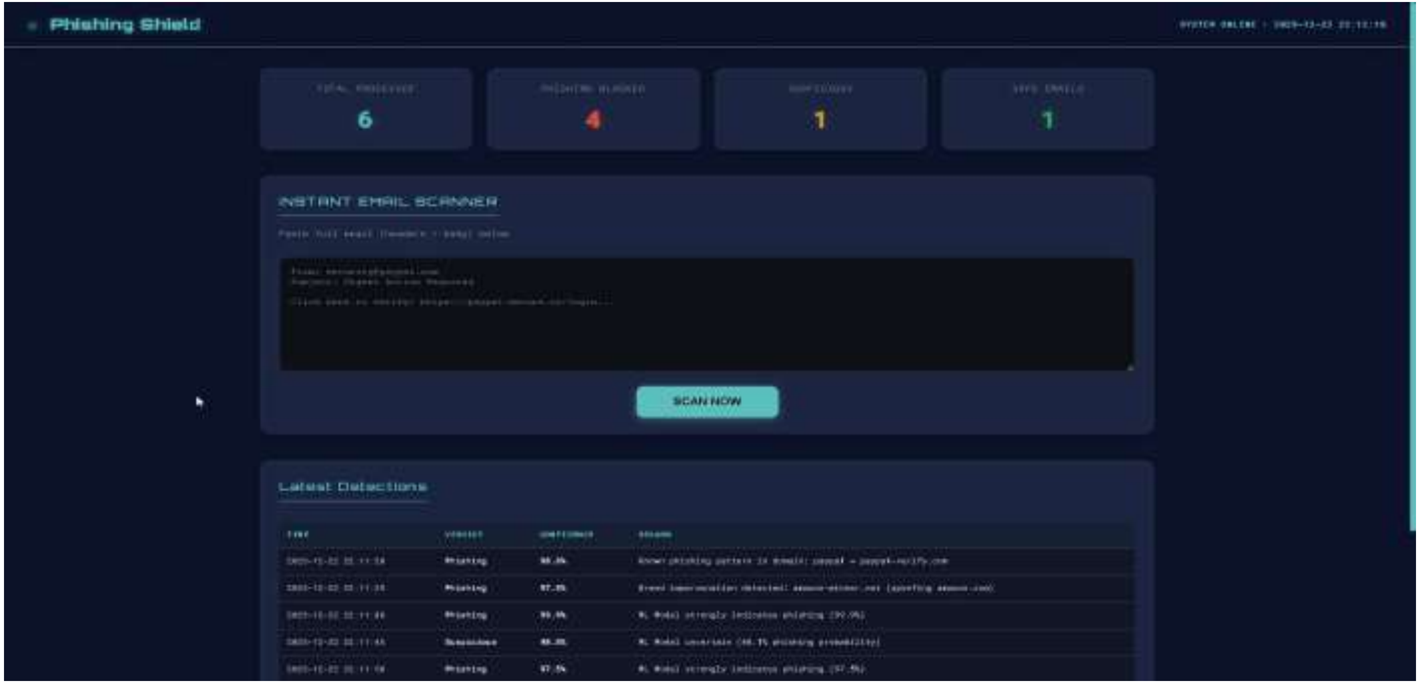
Feature View	**Model**	**What It Detects**
Word TF-IDF	Logistic Regression	Suspicious phrases & wording
Character TF-IDF	Linear SVM	Obfuscation & misspellings
Word Counts	Naive Bayes	Frequent phishing terms
Text Statistics	Random Forest	Style patterns (ALL CAPS, links, numbers)

Training & Validation Strategy

- Trained on **multiple datasets** for diversity
- Tested on a **completely unseen dataset**
- Uses cross-validation to avoid overfitting
- Final model retrained on all data for deployment
- Achieved **98.55% accuracy** on unseen data

Prototype

Prototype


Phishing Shield
6
4
1
1
INSTANT EMAIL SCANNER
SCAN NOW
Latest Detections

Any Questions?